PES UNIVERSITY

Department of Computer Science & Engineering

Digital Forensics and Incident Responses

UE23CS342BA6

Assignment 3B Submission

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Section	G
Department	CSE
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Topic: Lab 3B: File Analysis – EXIF data from Graphics Files

Digital Forensics and Incident Responses

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Task: Run Exiftool for the images and answer the questions

1. Unzip the Photos_Example folder with password “hands-on” and run exiftool

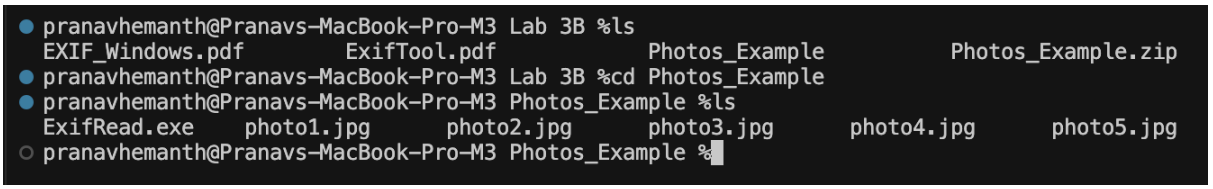


photo1.jpg:

```
pranavhemanth@Pranavs-MacBook-Pro-M3 Photos_Example %exiftool photo1.jpg
ExifTool Version Number         : 13.36
File Name                       : photo1.jpg
Directory                       : .
File Size                       : 3.9 MB
File Modification Date/Time     : 2015:10:23 13:38:00+05:30
File Access Date/Time           : 2026:02:13 10:45:37+05:30
File Inode Change Date/Time     : 2026:02:13 10:45:36+05:30
File Permissions                : -rw-rw-r--
File Type                       : JPEG
File Type Extension             : jpg
MIME Type                       : image/jpeg
Exif Byte Order                 : Big-endian (Motorola, MM)
Make                            : Apple
Camera Model Name               : iPhone 5s
Orientation                     : Horizontal (normal)
X Resolution                    : 72
Y Resolution                    : 72
Resolution Unit                 : inches
Software                        : 7.1.1
Modify Date                     : 2014:06:07 16:47:07
Y Cb Cr Positioning             : Centered
Exposure Time                   : 1/838
F Number                        : 2.2
Exposure Program                : Program AE
ISO                             : 32
Exif Version                    : 0221
Date/Time Original              : 2014:06:07 16:47:07
Create Date                     : 2014:06:07 16:47:07
Components Configuration        : Y, Cb, Cr, -
Shutter Speed Value             : 1/838
Aperture Value                  : 2.2
Brightness Value                : 8.689928058
Metering Mode                   : Multi-segment
Flash                           : Off, Did not fire
Focal Length                    : 4.1 mm
Subject Area                    : 1629 1222 1661 996
Maker Note Version              : 0
Run Time Flags                  : Valid
Run Time Value                  : 419651167477958
Run Time Scale                  : 1000000000
Run Time Epoch                  : 0
AE Stable                       : No
AE Target                       : 147
```

```
AE Average                      : 147
AF Stable                       : Yes
Sub Sec Time Original           : 103
Sub Sec Time Digitized          : 103
Flashpix Version                : 0100
Color Space                     : sRGB
Exif Image Width                : 3264
Exif Image Height               : 2448
Sensing Method                  : One-chip color area
Scene Type                      : Directly photographed
Exposure Mode                   : Auto
White Balance                   : Auto
Focal Length In 35mm Format     : 30 mm
Scene Capture Type              : Standard
Lens Info                       : 4.12mm f/2.2
Lens Make                       : Apple
Lens Model                      : iPhone 5s back camera 4.12mm f/2.2
GPS Latitude Ref                : North
GPS Longitude Ref               : West
GPS Altitude Ref                : Above Sea Level
GPS Time Stamp                  : 20:47:06.71
GPS Img Direction Ref           : True North
GPS Img Direction               : 216.1609756
Compression                     : JPEG (old-style)
Thumbnail Offset                : 1826
Thumbnail Length                : 12516
Image Width                     : 3264
Image Height                    : 2448
Encoding Process                : Baseline DCT, Huffman coding
Bits Per Sample                 : 8
Color Components                : 3
Y Cb Cr Sub Sampling            : YCbCr4:2:0 (2 2)
Run Time Since Power Up         : 4 days 20:34:11
Aperture                        : 2.2
Image Size                      : 3264x2448
Megapixels                      : 8.0
Scale Factor To 35 mm Equivalent: 7.3
Shutter Speed                   : 1/838
Create Date                     : 2014:06:07 16:47:07.103
Date/Time Original              : 2014:06:07 16:47:07.103
Thumbnail Image                 : (Binary data 12516 bytes, use -b option to extract)
GPS Altitude                    : 278.6 m Above Sea Level
GPS Latitude                    : 42 deg 43' 37.20" N
GPS Longitude                   : 78 deg 48' 49.48" W
Circle Of Confusion             : 0.004 mm
Field Of View                   : 61.9 deg
Focal Length                    : 4.1 mm (35 mm equivalent: 30.0 mm)
GPS Position                    : 42 deg 43' 37.20" N, 78 deg 48' 49.48" W
Hyperfocal Distance             : 1.87 m
Light Value                     : 13.6
Lens ID                         : iPhone 5s back camera 4.12mm f/2.2
pranavhemanth@Pranavs-MacBook-Pro-M3 Photos_Example %
```

photo2.jpg:

```
pranavhemanth@Pranavs-MacBook-Pro-M3 Photos_Example %exiftool photo2.jpg
ExifTool Version Number         : 13.36
File Name                       : photo2.jpg
Directory                       : .
File Size                       : 4.5 MB
File Modification Date/Time     : 2015:10:23 13:38:00+05:30
File Access Date/Time           : 2026:02:13 10:45:37+05:30
File Inode Change Date/Time     : 2026:02:13 10:45:36+05:30
File Permissions                : -rw-rw-r--
File Type                       : JPEG
File Type Extension             : jpg
MIME Type                       : image/jpeg
Exif Byte Order                 : Big-endian (Motorola, MM)
Make                            : Apple
Camera Model Name               : iPhone 4S
Orientation                     : Horizontal (normal)
X Resolution                    : 72
Y Resolution                    : 72
Resolution Unit                 : inches
Software                        : 5.0
Modify Date                     : 2011:08:25 09:27:36
Y Cb Cr Positioning             : Centered
Exposure Time                   : 1/1104
F Number                        : 2.4
Exposure Program                : Program AE
ISO                             : 64
Exif Version                    : 0221
Date/Time Original              : 2011:08:25 09:27:36
Create Date                     : 2011:08:25 09:27:36
Components Configuration        : Y, Cb, Cr, -
Shutter Speed Value             : 1/1104
Aperture Value                  : 2.4
Brightness Value                : 9.172699849
Metering Mode                   : Spot
Flash                           : Off, Did not fire
Focal Length                    : 4.3 mm
Subject Area                    : 1806 1008 610 612
Flashpix Version                : 0100
Color Space                     : sRGB
Exif Image Width                : 3264
Exif Image Height               : 2448
Sensing Method                  : One-chip color area
Exposure Mode                   : Auto
White Balance                   : Auto
Focal Length In 35mm Format     : 35 mm
Scene Capture Type              : Standard
Sharpness                       : Normal
GPS Latitude Ref                : North
GPS Longitude Ref               : West
GPS Altitude Ref                : Above Sea Level
GPS Time Stamp                  : 17:26:10
GPS Img Direction Ref           : True North
GPS Img Direction               : 254.5312024
Compression                     : JPEG (old-style)
Thumbnail Offset                : 908
Thumbnail Length                : 12028
Image Width                     : 3264
Image Height                    : 2448
Encoding Process                : Baseline DCT, Huffman coding
Bits Per Sample                 : 8
Color Components                : 3
```

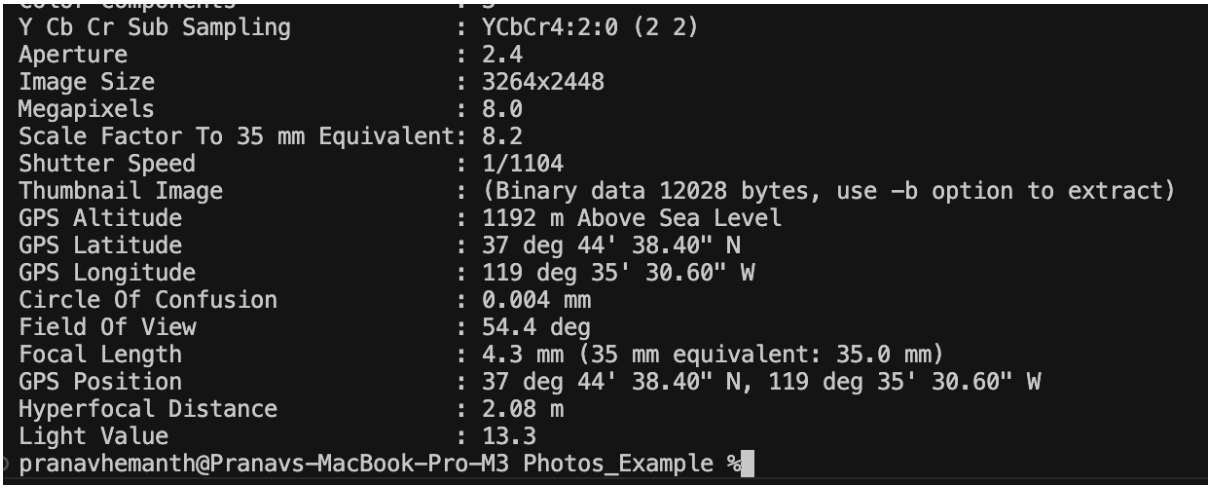

Y Cb Cr Sub Sampling : YCbCr4:2:0 (2 2)
Aperture : 2.4
Image Size : 3264x2448
Megapixels : 8.0
Scale Factor To 35 mm Equivalent: 8.2
Shutter Speed : 1/1104
Thumbnail Image : (Binary data 12028 bytes, use -b option to extract)
GPS Altitude : 1192 m Above Sea Level
GPS Latitude : 37 deg 44' 38.40" N
GPS Longitude : 119 deg 35' 30.60" W
Circle Of Confusion : 0.004 mm
Field Of View : 54.4 deg
Focal Length : 4.3 mm (35 mm equivalent: 35.0 mm)
GPS Position : 37 deg 44' 38.40" N, 119 deg 35' 30.60" W
Hyperfocal Distance : 2.08 m
Light Value : 13.3
pranavhemanth@Pranavs-MacBook-Pro-M3 Photos_Example %

photo3.jpg:

```
pranavhemanth@Pranavs-MacBook-Pro-M3 Photos_Example %exiftool photo3.jpg
ExifTool Version Number         : 13.36
File Name                       : photo3.jpg
Directory                       : .
File Size                       : 1011 kB
File Modification Date/Time     : 2015:10:23 13:38:00+05:30
File Access Date/Time           : 2026:02:13 10:45:37+05:30
File Inode Change Date/Time     : 2026:02:13 10:45:36+05:30
File Permissions                : -rw-rw-r--
File Type                       : JPEG
File Type Extension             : jpg
MIME Type                       : image/jpeg
Exif Byte Order                 : Little-endian (Intel, II)
Image Description               : Taken with my iPhone 5.
Make                            : Apple
Camera Model Name               : iPhone 5
X Resolution                    : 240
Y Resolution                    : 240
Resolution Unit                 : inches
Software                        : Adobe Photoshop Lightroom 5.0 (Windows)
Modify Date                     : 2013:10:20 10:43:35
Artist                          : Joe deSousa
Copyright                       : © 2013 Joe deSousa
Exposure Time                   : 1/1700
F Number                        : 2.4
Exposure Program                : Program AE
ISO                             : 80
Exif Version                    : 0230
Date/Time Original              : 2013:10:20 08:21:29
Create Date                     : 2013:10:20 08:21:29
Shutter Speed Value             : 1/1700
Aperture Value                  : 2.4
Brightness Value                : 9.435304199
Metering Mode                   : Multi-segment
Flash                           : On, Fired
Focal Length                    : 4.1 mm
Subject Area                    : 1631 1223 1795 1077
Sub Sec Time Original           : 186
Sub Sec Time Digitized          : 186
Color Space                     : sRGB
Sensing Method                  : One-chip color area
Scene Type                      : Directly photographed
Exposure Mode                   : Auto
White Balance                   : Auto
Focal Length In 35mm Format     : 33 mm
Scene Capture Type              : Standard
Lens Info                       : 4.12mm f/2.4
Lens Make                       : Apple
Lens Model                      : iPhone 5 back camera 4.12mm f/2.4
GPS Latitude Ref                : North
GPS Longitude Ref               : West
GPS Altitude Ref                : Above Sea Level
GPS Time Stamp                  : 12:21:28.36
GPS Img Direction Ref           : True North
GPS Img Direction               : 93.49529781
GPS Date Stamp                  : 2013:10:20
Compression                     : JPEG (old-style)
Thumbnail Offset                : 1124
Thumbnail Length                : 16965
Displayed Units X               : inches
Displayed Units Y               : inches
```

```
Current IPTC Digest             : e32ee593778c28eb8c1e501c6b6567bc
Coded Character Set             : UTF8
Application Record Version      : 4
Object Name                     : Erindale Park, Mississauga
Keywords                        : 5, apple, iPhone
Time Created                    : 08:21:29
Digital Creation Date           : 2013:10:20
Digital Creation Time           : 08:21:29
By-line                         : Joe deSousa
Province-State                  : Ontario
Country-Primary Location Name   : Canada
Copyright Notice                : © 2013 Joe deSousa
Caption-Abstract                : Taken with my iPhone 5.
Copyright Flag                  : True
Photoshop Thumbnail             : (Binary data 16965 bytes, use -b option to extract)
IPTC Digest                     : e32ee593778c28eb8c1e501c6b6567bc
Profile CMM Type                : Linotronic
Profile Version                 : 2.1.0
Profile Class                   : Display Device Profile
Color Space Data                : RGB
Profile Connection Space        : XYZ
Profile Date Time               : 1998:02:09 06:49:00
Profile File Signature          : acsp
Primary Platform                : Microsoft Corporation
CMM Flags                       : Not Embedded, Independent
Device Manufacturer             : Hewlett-Packard
Device Model                    : sRGB
Device Attributes               : Reflective, Glossy, Positive, Color
Rendering Intent                : Perceptual
Connection Space Illuminant     : 0.9642 1 0.82491
Profile Creator                 : Hewlett-Packard
Profile ID                      : 0
Profile Copyright               : Copyright (c) 1998 Hewlett-Packard Company
Profile Description             : sRGB IEC61966-2.1
Media White Point               : 0.95045 1 1.08905
Media Black Point               : 0 0 0
Red Matrix Column               : 0.43607 0.22249 0.01392
Green Matrix Column             : 0.38515 0.71687 0.09708
Blue Matrix Column              : 0.14307 0.06061 0.7141
Device Mfg Desc                 : IEC http://www.iec.ch
Device Model Desc               : IEC 61966-2.1 Default RGB colour space - sRGB
Viewing Cond Desc               : Reference Viewing Condition in IEC61966-2.1
Viewing Cond Illuminant         : 19.6445 20.3718 16.8089
Viewing Cond Surround           : 3.92889 4.07439 3.36179
Viewing Cond Illuminant Type    : D50
Luminance                       : 76.03647 80 87.12462
Measurement Observer            : CIE 1931
Measurement Backing             : 0 0 0
Measurement Geometry            : Unknown
Measurement Flare               : 0.999%
Measurement Illuminant          : D65
Technology                      : Cathode Ray Tube Display
Red Tone Reproduction Curve     : (Binary data 2060 bytes, use -b option to extract)
Green Tone Reproduction Curve   : (Binary data 2060 bytes, use -b option to extract)
Blue Tone Reproduction Curve    : (Binary data 2060 bytes, use -b option to extract)
XMP Toolkit                     : Adobe XMP Core 5.5-c002 1.148022, 2012/07/15-18:06:45
Creator Tool                    : Adobe Photoshop Lightroom 5.0 (Windows)
Metadata Date                   : 2013:10:20 10:43:35-04:00
Lens                            : iPhone 5 back camera 4.12mm f/2.4
Date Created                    : 2013:10:20 08:21:29.186
```

```
City                              : Mississauga
State                             : Ontario
Country                           : Canada
Document ID                       : xmp.did:607bd486-7847-7541-a10e-1bc3ee624e52
Original Document ID              : D543457BC1FBD482E72D3E16F57028B6
Instance ID                       : xmp.iid:607bd486-7847-7541-a10e-1bc3ee624e52
Format                            : image/jpeg
Marked                            : True
Version                           : 8.2
Process Version                   : 6.7
Auto White Version                : 134348800
Incremental Temperature           : +7
Incremental Tint                  : 0
Saturation                        : +7
Sharpness                         : 0
Luminance Smoothing               : 0
Color Noise Reduction             : 0
Vignette Amount                   : 0
Shadow Tint                       : 0
Red Hue                           : 0
Red Saturation                    : 0
Green Hue                         : 0
Green Saturation                  : 0
Blue Hue                          : 0
Blue Saturation                   : 0
Vibrance                          : +28
Hue Adjustment Red                : 0
Hue Adjustment Orange             : 0
Hue Adjustment Yellow             : 0
Hue Adjustment Green              : 0
Hue Adjustment Aqua               : 0
Hue Adjustment Blue               : 0
Hue Adjustment Purple             : 0
Hue Adjustment Magenta            : 0
Saturation Adjustment Red         : 0
Saturation Adjustment Orange      : 0
Saturation Adjustment Yellow      : 0
Saturation Adjustment Green       : 0
Saturation Adjustment Aqua        : 0
Saturation Adjustment Blue        : 0
Saturation Adjustment Purple      : 0
Saturation Adjustment Magenta     : 0
Luminance Adjustment Red          : 0
Luminance Adjustment Orange       : 0
Luminance Adjustment Yellow       : 0
Luminance Adjustment Green        : 0
Luminance Adjustment Aqua         : 0
Luminance Adjustment Blue         : 0
Luminance Adjustment Purple       : 0
Luminance Adjustment Magenta      : 0
Split Toning Shadow Hue           : 0
Split Toning Shadow Saturation    : 0
Split Toning Highlight Hue        : 0
Split Toning Highlight Saturation: 0
Split Toning Balance              : 0
Parametric Shadows                : 0
Parametric Darks                  : 0
Parametric Lights                 : 0
Parametric Highlights             : 0
Parametric Shadow Split           : 25
```

```
Parametric Midtone Split        : 50
Parametric Highlight Split      : 75
Sharpen Radius                  : +1.0
Sharpen Detail                  : 25
Sharpen Edge Masking            : 0
Post Crop Vignette Amount       : -16
Post Crop Vignette Midpoint     : 50
Post Crop Vignette Feather      : 50
Post Crop Vignette Roundness    : 0
Post Crop Vignette Style        : Highlight Priority
Post Crop Vignette Highlight Contrast: 0
Grain Amount                    : 0
Lens Profile Enable             : 1
Lens Manual Distortion Amount   : 0
Perspective Vertical            : 0
Perspective Horizontal          : 0
Perspective Rotate              : 0.0
Perspective Scale               : 100
Perspective Aspect              : 0
Perspective Upright             : Off
Auto Lateral CA                 : 0
Exposure 2012                   : 0.00
Contrast 2012                   : 0
Highlights 2012                 : -2
Shadows 2012                    : -5
Whites 2012                     : 0
Blacks 2012                     : 0
Clarity 2012                    : +28
Defringe Purple Amount          : 0
Defringe Purple Hue Lo          : 30
Defringe Purple Hue Hi          : 70
Defringe Green Amount           : 0
Defringe Green Hue Lo           : 40
Defringe Green Hue Hi           : 60
Convert To Grayscale            : False
Tone Curve Name 2012            : Linear
Camera Profile                  : Embedded
Lens Profile Setup              : Custom
Lens Profile Name               : Adobe (Apple iPhone 5)
Lens Profile Filename           : iPhone 5.lcp
Lens Profile Digest             : 9D7C9B1F8CCE979E8281042FDD7E0EB0
Lens Profile Distortion Scale   : 100
Lens Profile Chromatic Aberration Scale: 100
Lens Profile Vignetting Scale   : 100
Upright Version                 : 134217728
Upright Center Mode             : 0
Upright Center Norm X           : 0.5
Upright Center Norm Y           : 0.5
Upright Focal Mode              : 0
Upright Focal Length 35mm       : 35
Upright Preview                 : False
Upright Transform Count         : 0
Has Settings                    : True
Has Crop                        : False
Already Applied                 : True
History Action                  : derived, saved
History Parameters              : saved to new location
History Instance ID             : xmp.iid:607bd486-7847-7541-a10e-1bc3ee624e52
History When                    : 2013:10:20 10:43:35-04:00
History Software Agent          : Adobe Photoshop Lightroom 5.0 (Windows)
```

```
History Changed                 : /
Derived From Document ID        : D543457BC1FBD482E72D3E16F57028B6
Derived From Original Document ID: D543457BC1FBD482E72D3E16F57028B6
Creator                         : Joe deSousa
Title                           : Erindale Park, Mississauga
Rights                          : © 2013 Joe deSousa
Subject                         : 5, apple, iPhone
Description                     : Taken with my iPhone 5.
Usage Terms                     : All Rights Reserved
Tone Curve PV2012               : 0, 0, 255, 255
Tone Curve PV2012 Red           : 0, 0, 255, 255
Tone Curve PV2012 Green         : 0, 0, 255, 255
Tone Curve PV2012 Blue          : 0, 0, 255, 255
Retouch Area Spot Type          : clone
Retouch Area Source State       : sourceAutoComputed
Retouch Area Method             : gaussian
Retouch Area Source X           : 0.907697
Retouch Area Offset Y           : 0.030669
Retouch Area Opacity            : 1.000000
Retouch Area Feather            : 1.000000
Retouch Area Seed               : +2
Retouch Area Mask What          : Mask/Paint
Retouch Area Mask Value         : 1.000000
Retouch Area Mask Radius        : 0.020004
Retouch Area Mask Flow          : 1.000000
Retouch Area Mask Center Weight : 0.500000
Retouch Area Mask Dabs          : d 0.855331 0.010147
Gradient Based Corr What        : Correction
Gradient Based Corr Amount      : 1.000000
Gradient Based Corr Active      : true
Gradient Based Corr Saturation  : 0.000000
Gradient Based Corr Sharpness   : 0.000000
Gradient Based Corr Brightness  : 0.000000
Gradient Based Corr Toning Hue  : 233.000000
Gradient Based Corr Toning Saturation: 0.000000
Gradient Based Corr Exposure 2012: 0.000000
Gradient Based Corr Contrast 2012: 0.000000
Gradient Based Corr Highlights 2012: 0.000000
Gradient Based Corr Shadows 2012: 0.000000
Gradient Based Corr Clarity 2012: 0.000000
Gradient Based Corr Luminance Noise: 0.000000
Gradient Based Corr Moire       : 0.000000
Gradient Based Corr Defringe    : 0.000000
Gradient Based Corr Temperature : -0.382090
Gradient Based Corr Tint        : 0.000000
Gradient Based Corr Mask What   : Mask/Gradient
Gradient Based Corr Mask Value  : 1.000000
Gradient Based Corr Mask Zero X : 0.472840
Gradient Based Corr Mask Zero Y : 0.342012
Gradient Based Corr Mask Full X : 0.472618
Gradient Based Corr Mask Full Y : 0.008410
DCT Encode Version              : 100
APP14 Flags 0                   : (none)
APP14 Flags 1                   : (none)
Color Transform                 : YCbCr
Image Width                     : 3264
Image Height                    : 2448
Encoding Process                : Baseline DCT, Huffman coding
Bits Per Sample                 : 8
Color Components                : 3
```

```
Color Transform                 : YCbCr
Image Width                     : 3264
Image Height                    : 2448
Encoding Process                : Baseline DCT, Huffman coding
Bits Per Sample                 : 8
Color Components                : 3
Y Cb Cr Sub Sampling            : YCbCr4:4:4 (1 1)
Aperture                        : 2.4
Image Size                      : 3264x2448
Megapixels                      : 8.0
Scale Factor To 35 mm Equivalent: 8.0
Shutter Speed                   : 1/1700
Create Date                     : 2013:10:20 08:21:29.186
Date/Time Original              : 2013:10:20 08:21:29.186
Thumbnail Image                 : (Binary data 16965 bytes, use -b option to extract)
GPS Altitude                    : 105.7 m Above Sea Level
GPS Date/Time                   : 2013:10:20 12:21:28.36Z
GPS Latitude                    : 43 deg 32' 52.84" N
GPS Longitude                   : 79 deg 39' 17.72" W
Date/Time Created               : 2013:10:20 08:21:29
Digital Creation Date/Time      : 2013:10:20 08:21:29
Circle Of Confusion             : 0.004 mm
Field Of View                   : 57.2 deg
Focal Length                    : 4.1 mm (35 mm equivalent: 33.0 mm)
GPS Position                    : 43 deg 32' 52.84" N, 79 deg 39' 17.72" W
Hyperfocal Distance             : 1.89 m
Light Value                     : 13.6
Lens ID                         : iPhone 5 back camera 4.12mm f/2.4
pranavhemanth@Pranavs-MacBook-Pro-M3 Photos_Example %
```

photo4.jpg:

```
pranavhemanth@Pranavs-MacBook-Pro-M3 Photos_Example %exiftool photo4.jpg
ExifTool Version Number         : 13.36
File Name                       : photo4.jpg
Directory                       : .
File Size                       : 1562 kB
File Modification Date/Time     : 2015:10:23 13:38:00+05:30
File Access Date/Time           : 2026:02:13 10:45:37+05:30
File Inode Change Date/Time     : 2026:02:13 10:45:36+05:30
File Permissions                : -rw-rw-r--
File Type                       : JPEG
File Type Extension             : jpg
MIME Type                       : image/jpeg
JFIF Version                    : 1.01
Exif Byte Order                 : Big-endian (Motorola, MM)
Make                            : Apple
Camera Model Name               : iPhone 5s
Orientation                     : Horizontal (normal)
X Resolution                    : 72
Y Resolution                    : 72
Resolution Unit                 : inches
Software                        : Photos 1.0.1
Modify Date                     : 2015:01:14 13:10:38
Exposure Time                   : 1/3165
F Number                        : 2.2
Exposure Program                : Program AE
ISO                             : 32
Exif Version                    : 0221
Date/Time Original              : 2015:01:14 13:10:38
Create Date                     : 2015:01:14 13:10:38
Components Configuration        : Y, Cb, Cr, -
Shutter Speed Value             : 1/3165
Aperture Value                  : 2.2
Brightness Value                : 11.20418848
Exposure Compensation           : 0
Metering Mode                   : Multi-segment
Flash                           : Off, Did not fire
Focal Length                    : 4.2 mm
Subject Area                    : 1633 1220 1795 1077
Maker Note Version              : 2
Run Time Flags                  : Valid
Run Time Value                  : 310208277713708
Run Time Epoch                  : 0
Run Time Scale                  : 1000000000
AE Stable                       : Yes
AE Target                       : 218
AE Average                      : 224
AF Stable                       : Yes
Acceleration Vector             : -0.9915424889 0.000391620911 0.003489003918
Sub Sec Time Original           : 331
Sub Sec Time Digitized          : 331
Flashpix Version                : 0100
Color Space                     : sRGB
Exif Image Width                : 3264
Exif Image Height               : 2448
Sensing Method                  : One-chip color area
Scene Type                      : Directly photographed
Exposure Mode                   : Auto
White Balance                   : Auto
Digital Zoom Ratio              : 2.158730159
Focal Length In 35mm Format     : 62 mm
Scene Capture Type              : Standard
```

```
Lens Info                       : 4.15mm f/2.2
Lens Make                       : Apple
Lens Model                      : iPhone 5s back camera 4.15mm f/2.2
XMP Toolkit                     : XMP Core 5.4.0
Creator Tool                    : Photos 1.0.1
Date Created                    : 2015:01:14 13:10:38
Current IPTC Digest             : d41d8cd98f00b204e9800998ecf8427e
IPTC Digest                     : d41d8cd98f00b204e9800998ecf8427e
Profile CMM Type                : Linotronic
Profile Version                 : 2.1.0
Profile Class                   : Display Device Profile
Color Space Data                : RGB
Profile Connection Space        : XYZ
Profile Date Time               : 1998:02:09 06:49:00
Profile File Signature          : acsp
Primary Platform                : Microsoft Corporation
CMM Flags                       : Not Embedded, Independent
Device Manufacturer             : Hewlett-Packard
Device Model                    : sRGB
Device Attributes               : Reflective, Glossy, Positive, Color
Rendering Intent                : Perceptual
Connection Space Illuminant     : 0.9642 1 0.82491
Profile Creator                 : Hewlett-Packard
Profile ID                      : 0
Profile Copyright               : Copyright (c) 1998 Hewlett-Packard Company
Profile Description             : sRGB IEC61966-2.1
Media White Point               : 0.95045 1 1.08905
Media Black Point               : 0 0 0
Red Matrix Column               : 0.43607 0.22249 0.01392
Green Matrix Column             : 0.38515 0.71687 0.09708
Blue Matrix Column              : 0.14307 0.06061 0.7141
Device Mfg Desc                 : IEC http://www.iec.ch
Device Model Desc               : IEC 61966-2.1 Default RGB colour space - sRGB
Viewing Cond Desc               : Reference Viewing Condition in IEC61966-2.1
Viewing Cond Illuminant         : 19.6445 20.3718 16.8089
Viewing Cond Surround           : 3.92889 4.07439 3.36179
Viewing Cond Illuminant Type    : D50
Luminance                       : 76.03647 80 87.12462
Measurement Observer            : CIE 1931
Measurement Backing             : 0 0 0
Measurement Geometry            : Unknown
Measurement Flare               : 0.999%
Measurement Illuminant          : D65
Technology                      : Cathode Ray Tube Display
Red Tone Reproduction Curve     : (Binary data 2060 bytes, use -b option to extract)
Green Tone Reproduction Curve   : (Binary data 2060 bytes, use -b option to extract)
Blue Tone Reproduction Curve    : (Binary data 2060 bytes, use -b option to extract)
Image Width                     : 3264
Image Height                    : 2448
Encoding Process                : Baseline DCT, Huffman coding
Bits Per Sample                 : 8
Color Components                : 3
Y Cb Cr Sub Sampling            : YCbCr4:2:0 (2 2)
Run Time Since Power Up         : 3 days 14:10:08
Aperture                        : 2.2
Image Size                      : 3264x2448
Megapixels                      : 8.0
Scale Factor To 35 mm Equivalent: 14.9
Shutter Speed                   : 1/3165
Create Date                     : 2015:01:14 13:10:38.331
```

```
Date/Time Original              : 2015:01:14 13:10:38.331
Circle Of Confusion             : 0.002 mm
Field Of View                   : 32.4 deg
Focal Length                    : 4.2 mm (35 mm equivalent: 62.0 mm)
Hyperfocal Distance             : 3.89 m
Light Value                     : 15.5
Lens ID                         : iPhone 5s back camera 4.15mm f/2.2
pranavhemanth@Pranavs-MacBook-Pro-M3 Photos_Example %
```

photo5.jpg:

```
pranavhemanth@Pranavs-MacBook-Pro-M3 Photos_Example %exiftool photo5.jpg
ExifTool Version Number         : 13.36
File Name                       : photo5.jpg
Directory                       : .
File Size                       : 795 kB
File Modification Date/Time     : 2015:10:23 13:38:00+05:30
File Access Date/Time           : 2026:02:13 10:45:37+05:30
File Inode Change Date/Time     : 2026:02:13 10:45:36+05:30
File Permissions                : -rw-rw-r--
File Type                       : JPEG
File Type Extension             : jpg
MIME Type                       : image/jpeg
JFIF Version                    : 1.01
Exif Byte Order                 : Big-endian (Motorola, MM)
Make                            : Apple
Camera Model Name               : iPhone 5s
Orientation                     : Horizontal (normal)
X Resolution                    : 72
Y Resolution                    : 72
Resolution Unit                 : inches
Software                        : Photos 1.0.1
Modify Date                     : 2015:09:30 11:25:58
Exposure Time                   : 1/217
F Number                        : 2.2
Exposure Program                : Program AE
ISO                             : 32
Exif Version                    : 0221
Date/Time Original              : 2015:09:30 11:25:58
Create Date                     : 2015:09:30 11:25:58
Components Configuration        : Y, Cb, Cr, -
Shutter Speed Value             : 1/217
Aperture Value                  : 2.2
Brightness Value                : 12.94450374
Exposure Compensation           : 0
Metering Mode                   : Multi-segment
Flash                           : Off, Did not fire
Focal Length                    : 4.2 mm
Subject Area                    : 1627 1214 1797 1074
Maker Note Version              : 4
Run Time Flags                  : Valid
Run Time Value                  : 28907455419875
Run Time Epoch                  : 0
Run Time Scale                  : 1000000000
AE Stable                       : Yes
AE Target                       : 241
AE Average                      : 235
AF Stable                       : Yes
Acceleration Vector             : -0.005473210077 -0.9186824498 -0.409043659
Image Capture Type              : ProRAW
Sub Sec Time Original           : 798
Sub Sec Time Digitized          : 798
Flashpix Version                : 0100
Color Space                     : sRGB
Exif Image Width                : 2448
Exif Image Height               : 3264
Sensing Method                  : One-chip color area
Scene Type                      : Directly photographed
Exposure Mode                   : Auto
White Balance                   : Auto
Digital Zoom Ratio              : 4.995918367
Focal Length In 35mm Format     : 145 mm
```

```
Scene Capture Type              : Standard
Lens Info                       : 4.15mm f/2.2
Lens Make                       : Apple
Lens Model                      : iPhone 5s back camera 4.15mm f/2.2
GPS Version ID                  : 2.2.0.0
GPS Latitude Ref                : North
GPS Longitude Ref               : West
GPS Map Datum                   : WGS-84
Compression                     : JPEG (old-style)
Thumbnail Offset                : 1248
Thumbnail Length                : 2593
XMP Toolkit                     : XMP Core 4.4.0-Exiv2
Creator Tool                    : Photos 1.0.1
Date Created                    : 2015:09:30 11:25:58
IPTC Digest                     : d41d8cd98f00b204e9800998ecf8427e
Profile CMM Type                : Linotronic
Profile Version                 : 2.1.0
Profile Class                   : Display Device Profile
Color Space Data                : RGB
Profile Connection Space        : XYZ
Profile Date Time               : 1998:02:09 06:49:00
Profile File Signature          : acsp
Primary Platform                : Microsoft Corporation
CMM Flags                       : Not Embedded, Independent
Device Manufacturer             : Hewlett-Packard
Device Model                    : sRGB
Device Attributes               : Reflective, Glossy, Positive, Color
Rendering Intent                : Perceptual
Connection Space Illuminant     : 0.9642 1 0.82491
Profile Creator                 : Hewlett-Packard
Profile ID                      : 0
Profile Copyright               : Copyright (c) 1998 Hewlett-Packard Company
Profile Description             : sRGB IEC61966-2.1
Media White Point               : 0.95045 1 1.08905
Media Black Point               : 0 0 0
Red Matrix Column               : 0.43607 0.22249 0.01392
Green Matrix Column             : 0.38515 0.71687 0.09708
Blue Matrix Column              : 0.14307 0.06061 0.7141
Device Mfg Desc                 : IEC http://www.iec.ch
Device Model Desc               : IEC 61966-2.1 Default RGB colour space - sRGB
Viewing Cond Desc               : Reference Viewing Condition in IEC61966-2.1
Viewing Cond Illuminant         : 19.6445 20.3718 16.8089
Viewing Cond Surround           : 3.92889 4.07439 3.36179
Viewing Cond Illuminant Type    : D50
Luminance                       : 76.03647 80 87.12462
Measurement Observer            : CIE 1931
Measurement Backing             : 0 0 0
Measurement Geometry            : Unknown
Measurement Flare               : 0.999%
Measurement Illuminant          : D65
Technology                      : Cathode Ray Tube Display
Red Tone Reproduction Curve     : (Binary data 2060 bytes, use -b option to extract)
Green Tone Reproduction Curve   : (Binary data 2060 bytes, use -b option to extract)
Blue Tone Reproduction Curve    : (Binary data 2060 bytes, use -b option to extract)
Image Width                     : 2448
Image Height                    : 3264
Encoding Process                : Baseline DCT, Huffman coding
Bits Per Sample                 : 8
Color Components                : 3
Y Cb Cr Sub Sampling            : YCbCr4:2:0 (2 2)
```

```
Run Time Since Power Up         : 8:01:47
Aperture                        : 2.2
Image Size                      : 2448x3264
Megapixels                      : 8.0
Scale Factor To 35 mm Equivalent: 34.9
Shutter Speed                   : 1/217
Create Date                     : 2015:09:30 11:25:58.798
Date/Time Original              : 2015:09:30 11:25:58.798
Thumbnail Image                 : (Binary data 2593 bytes, use -b option to extract)
GPS Latitude                    : 28 deg 16' 23.46" N
GPS Longitude                   : 81 deg 38' 52.32" W
Circle Of Confusion             : 0.001 mm
Field Of View                   : 14.2 deg
Focal Length                    : 4.2 mm (35 mm equivalent: 145.0 mm)
GPS Position                    : 28 deg 16' 23.46" N, 81 deg 38' 52.32" W
Hyperfocal Distance             : 9.10 m
Light Value                     : 11.7
Lens ID                         : iPhone 5s back camera 4.15mm f/2.2
pranavhemanth@Pranavs-MacBook-Pro-M3 Photos_Example %
```

Question: Answer below based on above information

ANSWER BELOW MENTIONED QUESTIONS WITH REGARD TO PHOTO 1:

1. Based on the EXIF data, when was the picture taken?
   Answer: The picture was taken on 2014:06:07 16:47:07.103

2. On what device was the picture taken?
   Answer: Apple iPhone 5s was used to take the picture

3. Was the picture taken with the front camera lens or the back camera lens?
   Answer: The picture was taken with the back camera (Lens Model: iPhone 5s back camera 4.12mm f/2.2)

4. What were the GPS coordinates of the camera/phone, when the picture was taken?
   Answer: The coordinates were 42° 43’ 37.20” N, 78° 48’ 49.48” W (Approx. 42.7270° N, -78.8137° W)

5. In a browser open http://www.google.com/maps and determine the location of the coordinates.
   Answer: The coordinates correspond to - Buffalo, New York, USA

ANSWER BELOW MENTIONED QUESTIONS WITH REGARD TO PHOTO 2:

6. When was photo2.jpg taken based on the EXIF data?
   Answer: The photo was taken on 2011:08:25 09:27:36

7. Where was the camera located when photo2.jpg was taken based on the EXIF data?
   Answer: Based on the exif data, the coordinates are - 37° 44’ 38.40” N, 119° 35’ 30.60” W (Approx. 37.7440° N, -119.5918° W) Location: Yosemite National Park, California, USA

8. Are there any indicators in the EXIF data, which suggest the picture or EXIF data in photo2.jpg were manipulated by a third-party tool?
   Answer: No. There are no indicators of third-party editing software in the EXIF data.

ANSWER BELOW MENTIONED QUESTIONS WITH REGARD TO PHOTO 3:

9. When was photo3.jpg taken based on the EXIF data?
   Answer: The photo was taken on 2013:10:20 08:21:29.186

10. Where was the camera located when photo3.jpg was taken based on the EXIF data?
    Answer: Based on the exif data, the coordinates are - 43° 32’ 52.84” N, 79° 39’ 17.72” W (Approx. 43.5480° N, -79.6549° W) Location: Erindale Park, Mississauga, Ontario, Canada

11. Are there any indicators in the EXIF data, which suggest the picture or EXIF data in photo3.jpg were manipulated by a third-party tool?
    Answer: Yes. The metadata shows Adobe Photoshop Lightroom 5.0 (Windows) and extensive XMP editing data, indicating post-processing.

ANSWER BELOW MENTIONED QUESTIONS WITH REGARD TO PHOTO 4:

12. When was photo4.jpg taken based on the EXIF data?
    Answer: The photo was taken on 2015:01:14 13:10:38.331

13. Where was the camera located when photo4.jpg was taken based on the EXIF data?
    Answer: No GPS latitude or longitude information is present in the EXIF data. Location cannot be determined.

14. Are there any indicators in the EXIF data, which suggest the picture or EXIF data in photo4.jpg were manipulated by a third-party tool?
    Answer: No third-party editing tools are indicated. The image was processed using Apple Photos 1.0.1 only.

ANSWER BELOW MENTIONED QUESTIONS WITH REGARD TO PHOTO 5:

15. When was photo5.jpg taken based on the EXIF data?
    Answer: The photo was taken on 2015:09:30 11:25:58.798

16. Where was the camera located when photo5.jpg was taken based on the EXIF data?
    Answer: Based on the exif data, the coordinates are - 28° 16’ 23.46” N, 81° 38’ 52.32” W (Approx. 28.2732° N, -81.6479° W) Location: Orlando, Florida, USA (near Walt Disney World area)

17. Are there any indicators in the EXIF data, which suggest the picture or EXIF data in photo5.jpg were manipulated by a third-party tool?
    Answer: No. Only Apple Photos 1.0.1 is listed. There are no indicators of third-party editing software.